

Data Analysis and Statistics

Effect of Global warming on rising
temperature.

Assignment 2- Report

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Introduction

The rise in temperature due to global warming has been an ongoing concern for the past two decades. Ever since the dawn of the 19th century there has been a booming increase in the mean global temperature and global warming has been deemed to be responsible for it. The phenomenon of the natural greenhouse effect is responsible for the earth to maintain its temperature which is suitable for all living things to thrive (Kweku, D. W. et al, 2018). The natural greenhouse effect relates to the heat trapped in the earth's atmosphere due to the greenhouse gasses present which helps maintain a comfortable global mean temperature. However, with the rise in these greenhouse gasses the increase in heat trapped inside the atmosphere has increased in the last three decades (Khandekar, M. et al, 2005).

■ Case study and Importance

This case study is to identify the trends in the increase in the Greenhouse gasses and its effects on the mean global temperature of the earth. It is of high importance to study and observe the climatic changes that are being brought about by human activities. The deterioration of the earth's atmosphere affects every living organism and in turn whole ecosystems. It is important to take the necessary steps to reduce the ill effects caused on the atmosphere by human activities. By observing and studying the trends of the previous decades we can predict the future challenges that might be faced on a global scale.

■ Challenges identified

There are a number of issues that result in the rise of Global warming and consecutively the mean Global Temperature of the planet (Mohammed, Y. S. et al, 2012) They are as follows,

1. Interaction of solar radiation with the greenhouse gasses present in the atmosphere.
2. Cutting down of forests.
3. Manufacturing goods on a large scale.
4. Generating electricity and heat by burning of fossil fuels.
5. Increase in use of private transportation.
6. Increase in global urbanization and population.

- Data Selection and Processing:

- Data Selection

The dataset has been taken from Berkeley Earth which is affiliated with Lawrence Berkeley National Laboratory.

The logo for Berkeley Earth, featuring a blue circular graphic with a white center, and the text "BERKELEY EARTH" in a bold, sans-serif font, with a trademark symbol (TM) to the right.

The Dataset contains:

- Mean temperatures of various different cities and countries.
- Data is divided into regions for better understanding of their contribution in the overall global temperature.
- Maps the average temperatures yearly from 1995 – 2020 with a gap of five years (1995,2000,2005,2010,2015,2020).
- The percent contributions of each country to the global level of greenhouse gasses are also present in the dataset.

▪ Data Pre-processing:

1. The data had consisted of null values which were removed and replaced with '0' in case no data was present.
2. Redundant data that is duplicate rows and columns were removed.
3. The regions column was added to properly identify which country belonged to the respective continent.
4. The percentage of greenhouse gasses each country has contributed over the span of years are added to the dataset.
5. The dataset was reduced to pick out the major cities within the regions spread out in the geographical layout of the continent to get precise values about the temperature of each region.
6. Yearly data over a span of five years observed in January is taken into consideration in this dataset.

Obs	Region	Country	City	Month	Day	Year	AvgTemperature	Levelofglobaltotal
1	Africa	Algeria	Algiers	1	29	1995	54.3	0.40%
2	Africa	Algeria	Algiers	1	29	2000	50.3	0.40%
3	Africa	Algeria	Algiers	1	29	2005	43.8	0.40%
4	Africa	Algeria	Algiers	1	29	2010	51.1	0.40%
5	Africa	Algeria	Algiers	1	29	2015	53.6	0.41%
6	Africa	Algeria	Algiers	1	29	2020	55	0.45%
7	Africa	Madagascar	Antananarivo	1	29	1995	70.7	0.05%
8	Africa	Madagascar	Antananarivo	1	29	2000	70.4	0.05%
9	Africa	Madagascar	Antananarivo	1	29	2005	67.5	0.05%
10	Africa	Madagascar	Antananarivo	1	29	2010	74	0.09%

7. The above data represents 10 observations from the dataset which is selected.
8. The Categorical variables are Region, Country and City.
9. The Continuous Variables include Month, Day, AvgTemperature and Levelofglobaltotal.
10. Here, Year can be considered as an ordinal variable as it is divided into a hierarchy of 1995,2000,2005,2010,2015 and 2020 respectively.

- Proposed solutions and justifications
 - Analytical methods
 - Descriptive Analysis (What is the problem?):

The amount of CO₂ which is a greenhouse gas that helps trap the heat in the atmosphere by has increased by 50% since the Industrial Revolution began. This sharp rise in CO₂ since the global industrialization is the most important climate change driver over the last century. The rise in CO₂ emissions directly affects and leads to the rise in the temperature of the respective region. The increase in the levels of CO₂ in a particular region leads to the increase in the effects of global warming.

- Diagnostic Analysis (Why is this problem occurring?):

Greenhouse gas concentrations have increased due to human activity. Burning fossil fuels like coal and oil has increased the amount of carbon dioxide in the atmosphere during the past century (CO₂). This rise is brought on by the burning of coal or oil, which produces CO₂ when airborne carbon and oxygen mix. The natural greenhouse affect causes the heat emitted by the earth in longer wavelengths to get absorbed by the greenhouse gasses (water vapour, carbon dioxide, etc.) in the atmosphere. This helps regulate the annual temperature of the earth. However, with the increase of the carbon dioxide which is a greenhouse gas. The amount of heat getting absorbed and trapped in the earth's atmosphere has increased. Thus, leading to the temperature rise and as a result global warming.

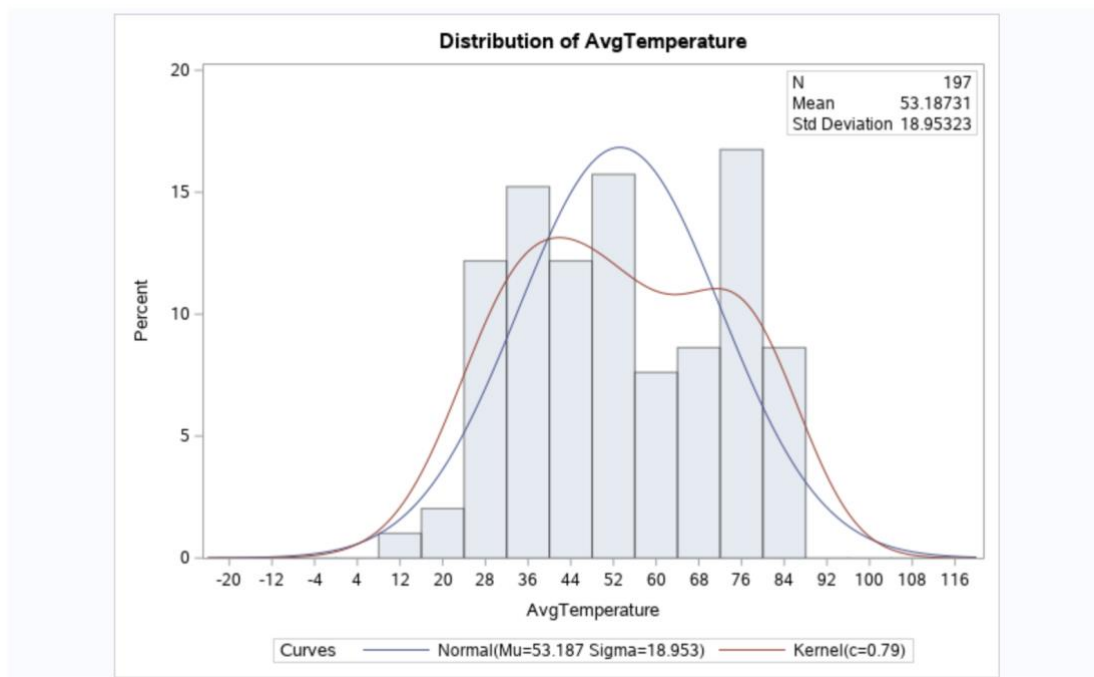
■ Correlational Analysis:

Various statistical methods can be used to discover if there is a relationship between two variables(predictor/response), and how strong that relationship may be.

Type of Predictors Type of Response	Categorical	Continuous	Continuous and Categorical
Continuous	Analysis of Variance (ANOVA)	Ordinary Least Squares (OLS) Regression	Analysis of Covariance (ANCOVA)
Categorical	Logistic Regression	Logistic Regression	Logistic Regression

Normal/Gaussian Distribution:

Before we go ahead with the analysis of the statistical methods the below graph shows the normal/Gaussian distribution of average temperature with 95% confidence. The Number of datapoints in the dataset is 197. The mean of the population data is 53.18 and the standard deviation is 18.95.



▪ Statistical Methods:

1. Hypothesis Testing Analysis:

1.1. T- tests

⇒ One -Sample t-test.

○ NULL Hypothesis(H_0):

THERE IS NO CHANGE IN THE GLOBAL TEMPERATURE WITH THE INCREASE OF GREENHOUSE GASSES.

○ ALTERNATIVE Hypothesis(H_1):

THERE IS A CHANGE IN THE GLOBAL TEMPERATURE WITH THE INCREASE OF GREENHOUSE GASSES.

$p \geq 0.05$ Do Not Reject H_0

$p < 0.05$ Reject H_0

Using the one -sample t -test with the average temperature as the data the p value is <0.001 .

Therefore, we can successfully reject the null (H_0) hypothesis. Thus, the alternative hypothesis(H_1) is true.

Variable: AvgTemperature

Tests for Location: $\mu_0=0$				
Test	Statistic		p Value	
Student's t	t	39.38745	Pr > t	<.0001
Sign	M	98.5	Pr >= M	<.0001
Signed Rank	S	9751.5	Pr >= S	<.0001

⇒ Two -sample t-test:

- NULL Hypothesis(H_0):
THE MEAN TEMPERATURES FOR ASIA AND AFRICA ARE THE SAME
- ALTERNATIVE Hypothesis(H_1):
THE MEAN TEMPERATURES FOR ASIA AND AFRICA ARE NOT THE SAME.

$p \geq 0.05$ Do Not Reject H_0
 $p < 0.05$ Reject H_0

Here, using the two -sample t- test with region and average temperature, the below data was obtained. Since the p value < 0.05 we can reject the null hypothesis. Thus, concluding the alternative hypothesis as true.

Wilcoxon Scores (Rank Sums) for Variable AvgTemperature Classified by Variable Region					
Region	N	Sum of Scores	Expected Under H_0	Std Dev Under H_0	Mean Score
Africa	54	7648.50	5346.0	356.944534	141.638889
Asia	42	4570.00	4158.0	327.737647	108.809524
Austra	18	2553.50	1782.0	230.567773	141.861111
Europe	83	4731.00	8217.0	395.118449	57.000000
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
86.6651	3	<.0001

Using the **Chi -square** test along with the hypothesis testing we have concluded that there does exist some relation between the two variables although we don't know how strong the relation is.

1.2. F- tests

A Statistical F -Test uses an F statistic to compare two variances.

Below is the formula used to calculate the F statistic

$$\text{F-Statistic: } F = \frac{\max(s_1^2, s_2^2)}{\min(s_1^2, s_2^2)}$$

The below table shows the F statistic value to be 2.72.

Effect Entered: Levelofglobaltotal

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	92	49758	540.84496	2.72	<.0001
Error	104	20650	198.56118		
Corrected Total	196	70408			

If the F value is larger, it proves that it goes against the assumption of equality. Thus, we have a p value which is <0.05 which is the assumption we have considered for ALPHA. Also, the F statistic value proves that there is a strong relation between the two variables and hence, the alternative hypothesis is true.

2. (OLS) Ordinary Least Square regression

The relationship between one or more independent quantitative variables and a dependent variable is described by the OLS regression test.

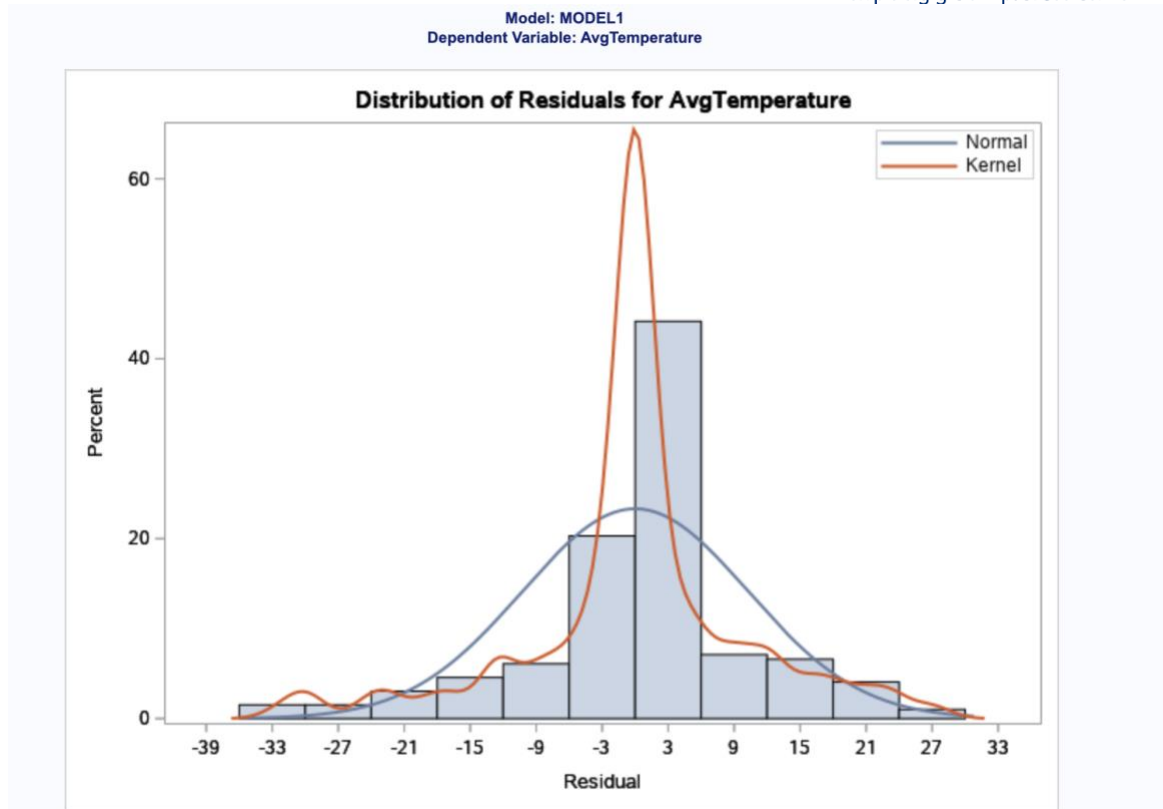
The unknown parameters in a model are estimated using this technique, which is a type of linear regression. The approach is based on reducing the sum of squared residuals between the actual and anticipated values.

“How much does the increase in the levels of greenhouse gasses affect the temperature rise over a span of 25 years?”

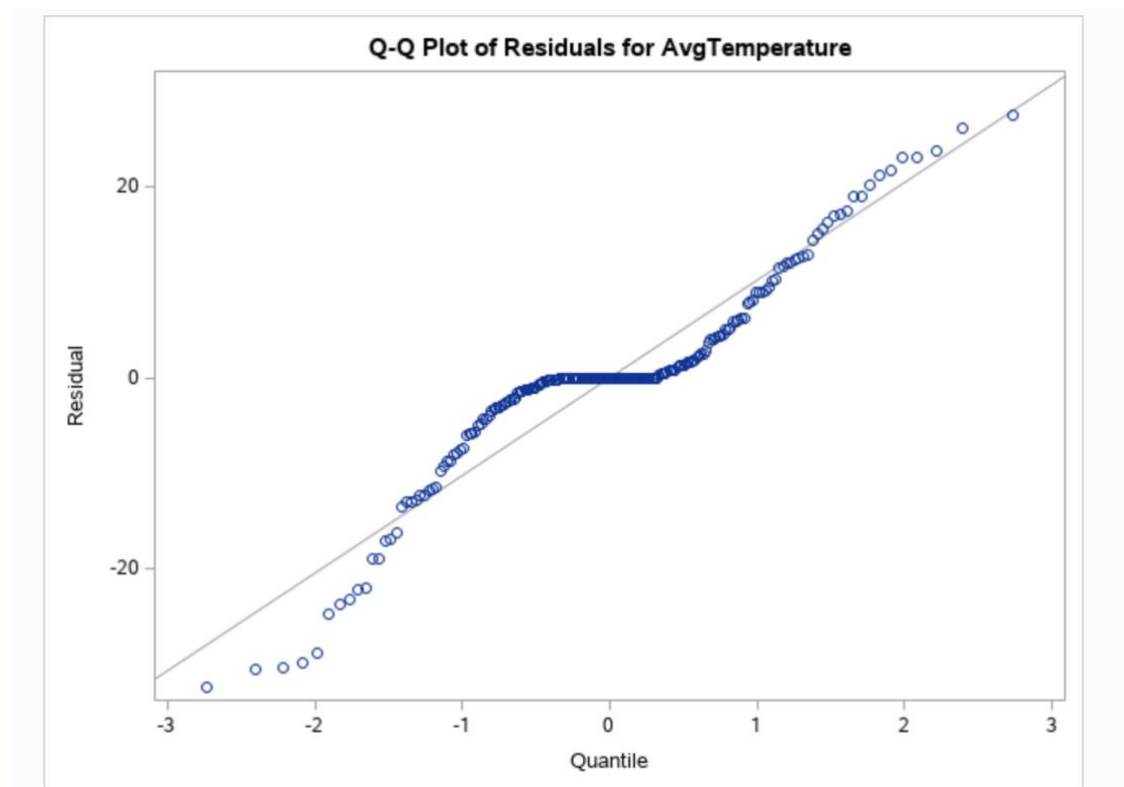
In the table below, the value of R- square is 0.7067 which is closer to 1. This shows that the variables used in this case that are AvgTemperature and Levelofglobaltotal are significantly related to each other. A change in the value of one variable will cause a change in the other.

Root MSE	14.09117
Dependent Mean	53.18731
R-Square	0.7067
Adj R-Sq	0.4473
AIC	1301.50003
AICC	1476.59807
SBC	1407.83798

The graph below shows the increase of global greenhouse gasses causes a spike in the average temperature percent. There is a significant rise of about 60% in the temperature if the value of the greenhouse gasses varies between -3 to 3.



Here, the below graph shows some skewness towards the ends where the outliers may be present. Most of the data lies between 0 and 1.



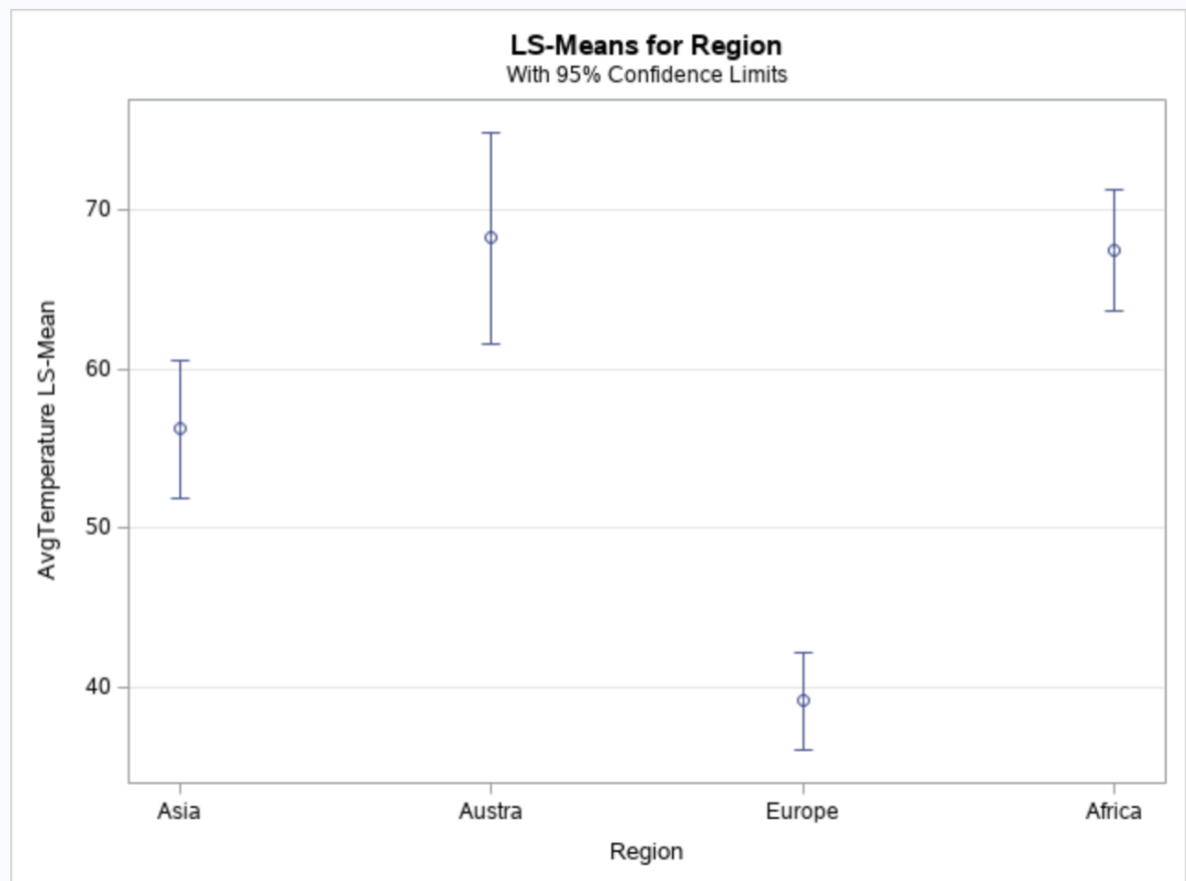
3. ANCOVA (Analysis of Covariance)

The ANCOVA model combines ANOVA and Regression to form a General Linear Model. Below is the question proposed to solve.

“How many regions have contributed to the increase in mean temperature of $>40.6^{\circ}\text{F}$ since the 1995?”

R-Square	Coeff Var	Root MSE	AvgTemperature Mean
0.455042	26.78884	14.24827	53.18731

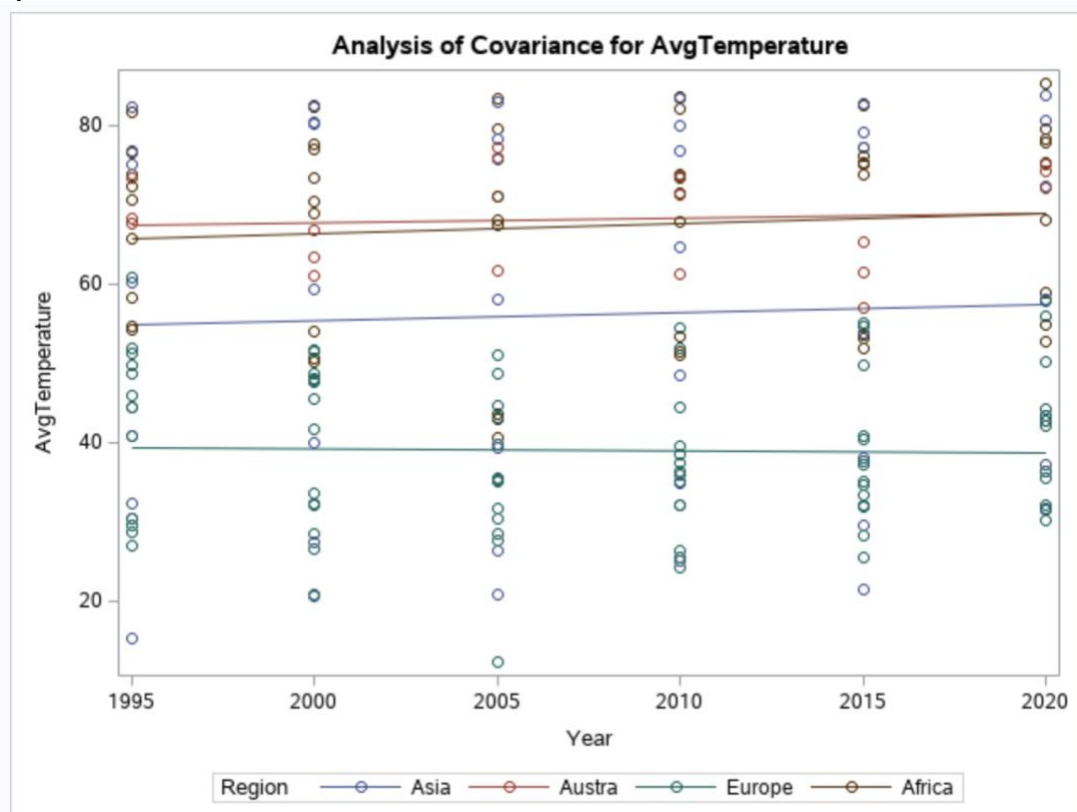
Here, the average temperature mean calculated from 1995-2020 is 53.18°F as shown in the table above. As represented in the table and graph below, the three regions which have contributed to the increase in the average temperature since 1995 are Australia, Africa and Asia.



**Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer**

Region	AvgTemperature LSMEAN	LSMEAN Number
Asia	56.2227473	1
Austra	68.2451793	2
Europe	39.1016485	3
Africa	67.4608482	4

The table shows the average temperature of Africa and Australia to have increased to 67.46 °F and 68.24 °F which has caused the increase in the average temperature mean globally. The below graph shows the variance and the relation between the average temperature and year. It also helps observe the slow but gradual increase in the temperatures of Africa and Australia respectively. There also shows a spike in the increase of the mean temperature of Asia which is 56.22°F. However, this has not contributed as much to the overall global average temperature.



Thus, using the ANCOVA method and finding the variance in the temperature mean over the years we can conclude how many regions have contributed to the increase of the global temperature mean.

4. Logistic regression.

A logistic regression model presents the data and explains how one dependent binary variable and one or more independent nominal, ordinal, interval, or ratio-level variables relate to one another.

“Will the rise in greenhouse gasses lead to an increase in rate of Global warming?”

The below data was obtained using the dataset and comparing a continuous predictor (avgtemperature) to another categorical variable (region).

Model Information	
Data Set	TEMP.CITY_TEMP
Response Variable	Region
Number of Response Levels	4
Model	cumulative logit
Optimization Technique	Fisher's scoring

Number of Observations Read	197
Number of Observations Used	197

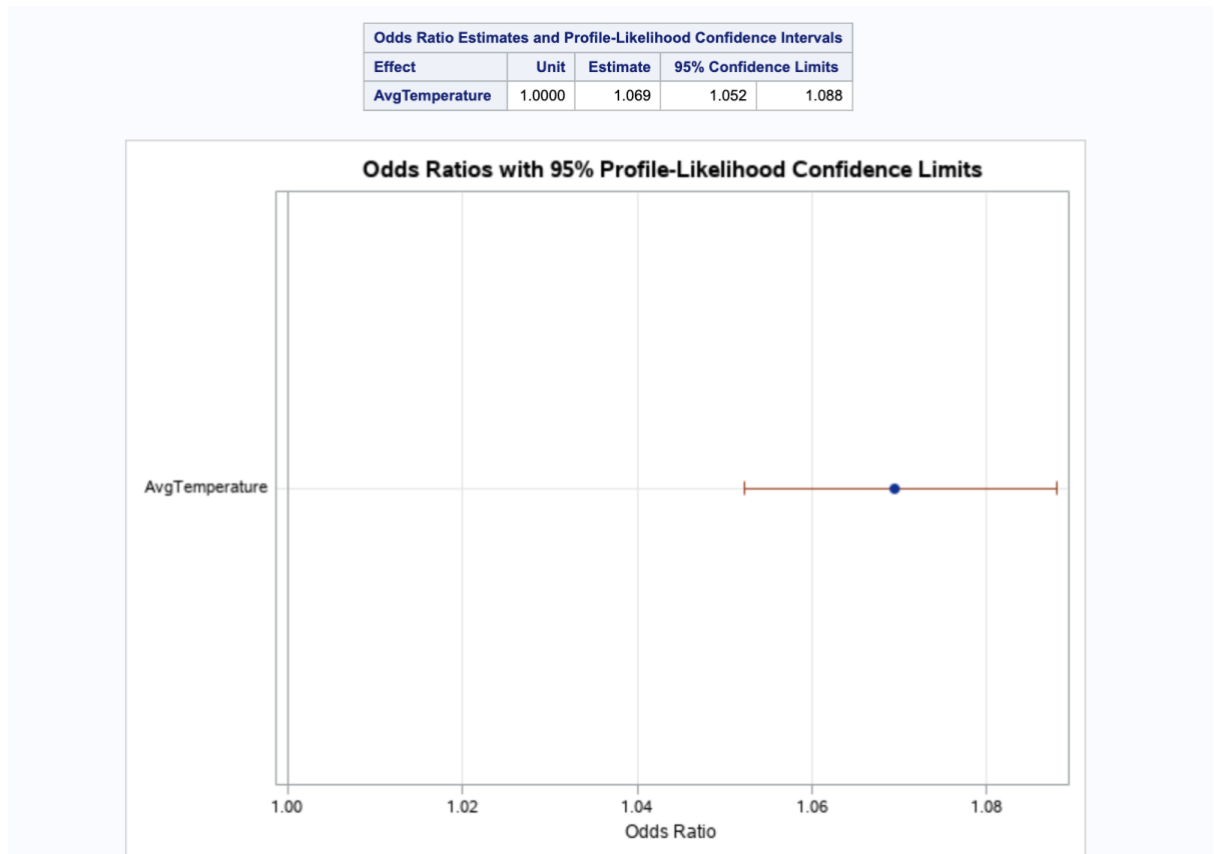
Response Profile		
Ordered Value	Region	Total Frequency
1	Africa	54
2	Asia	42
3	Austra	18
4	Europe	83

Probabilities modeled are cumulated over the lower Ordered Values.

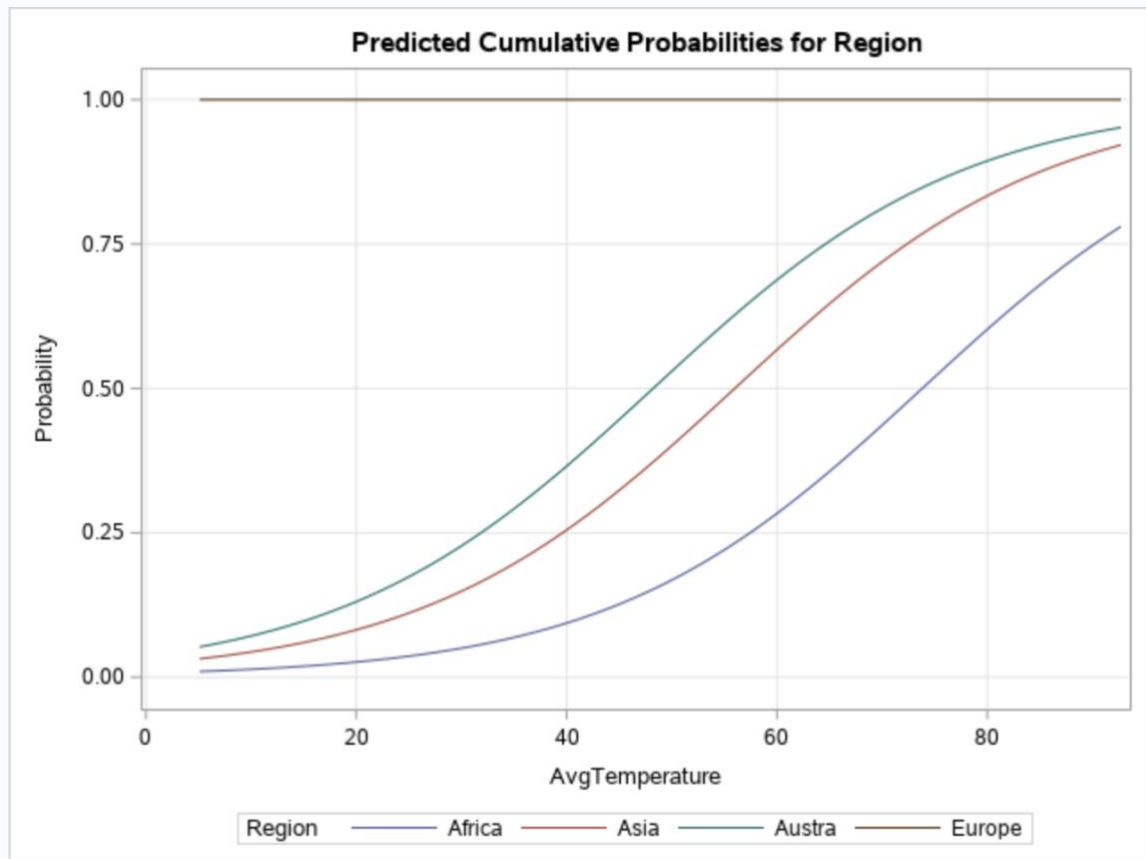
Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
92.7301	2	<.0001

We have previously concluded that the regions of Australia, Africa and Asia play a big role in the increase of greenhouse gasses. Keeping that in mind we can check the probability of rise in the temperature due to an increase in the same.



The above graph uses the odds ratio estimates and the Profile-Likelihood to find the probability of temperature increase with the increase with the greenhouse gasses of the regions. The graph shows that the estimated value with regard to 95% confidence region is from 1.052 to 1.088. Here, the increase in temperature for 1°F is estimated every 1.069 units. Thus, the true probability value lies between these two values.



The above graph shows the probability of temperature increase. The probability of increase in temperature increases as we go from the west to the east in terms of location. The levels of greenhouse gasses have increased more due to the Australian, African and Asian regions as concluded in the previous model. Hence, this graph proves that the increase in CO₂ increases the probability of temperature rise of the region. And as a result, the average temperature mean of global temperature further facilitating global warming.

• Conclusion

- By using the dataset provided by BERKELY EARTH, it is possible to find the trends in the increase in the greenhouse gas (CO₂) which in turn causes an increase in temperature. The findings can be supported by graphical analysis and hypothesis testing, with 95% Confidence Intervals throughout. Various statistical methods are used to answer the common questions that can be asked to the dataset as well as to find any trends that the data represents.
- The findings are visualized into a reader friendly format that allows for actionable decisions to be undertaken to carry out the measures to help reduce the overall global temperature.
- We were able to estimate the probability of increase in the temperature mean and also the dependence of the variables on one another.
- The approximate standard deviation and variance in the dataset could also be calculated using the statistical techniques provided by SAS.
- Using these statistical methods, it was also possible to carry out the hypothesis testing of the student t test and the F statistic test to agree with or rule out the statements prophesied.
- Hence, we can prove that there does exist a possible relation between the increase in global warming and temperature rise of the planet.

References:

- Khandekar, M. L., Murty, T. S., & Chittibabu, P. (2005). The global warming debate: A review of the state of science. *Pure and Applied Geophysics*, 162(8), 1557-1586.
- Kweku, D. W., Bismark, O., Maxwell, A., Desmond, K. A., Danso, K. B., Oti-Mensah, E. A., ... & Adormaa, B. B. (2018). Greenhouse effect: greenhouse gases and their impact on global warming. *Journal of Scientific research and reports*, 17(6), 1-9.
- Mohammed, Y. S., Mokhtar, A. S., Bashir, N., Abdullahi, U. U., Kaku, S. J., & Umar, U. (2012). A synopsis on the effects of anthropogenic greenhouse gases emissions from power generation and energy consumption. *International Journal of Scientific and Research Publications*, 2(10), 1-6.

Appendix:

- */* Code to obtain the first 10 observations of the dataset*/*

```
proc print data=temp.city_temp (obs=10);  
run;
```

- */* Distribution Analysis of Average temperature*/*

```
proc univariate data=TEMP.CITY_TEMP;  
    ods select Histogram;  
    var AvgTemperature Year;  
    histogram AvgTemperature Year / normal kernel;  
    inset n mean std / position=ne;  
run;
```

- */* one -sample t-test hypothesis testing*/*

```
proc ttest data=TEMP.CITY_TEMP sides=2 h0=0.05 plots(only  
showh0)=(summaryPlot  
            intervalPlot qqplot);  
    var AvgTemperature;  
run;  
  
/* Nonparametric test */  
proc univariate data=TEMP.CITY_TEMP mu0=0.05;  
    ods select TestsForLocation;  
    var AvgTemperature;  
run;
```

■ */* two -sample t-test hypothesis testing*/*

```
proc ttest data=TEMP.CITY_TEMP sides=2 h0=0 plots(only showh0)=(summaryPlot
    intervalPlot qqplot);
    class Region;
    var AvgTemperature;
```

```
run;
```

```
/* Nonparametric test */
```

```
proc npar1way data=TEMP.CITY_TEMP wilcoxon plots=none;
    class Region;
    var AvgTemperature;
```

```
run;
```

■ */* Ordinary Least Square Regression test for variance*/*

```
ods noproctitle;
ods graphics / imagemap=on;
```

```
proc glmselect data=TEMP.CITY_TEMP outdesign(addinputvars)=Work.reg_design
    plots=(criterionpanel(unpack) );
    class Levelofglobaltotal / param=glm;
    model AvgTemperature=Levelofglobaltotal / showpvalues selection=forward
```

```
(select=adjrsq choose=adjrsq) details=steps stats=(adjrsq rsquare);
run;
```

```
proc reg data=Work.reg_design alpha=0.05 plots(only)=(diagnostics(unpack)
    residuals observedbypredicted);
    where Levelofglobaltotal is not missing;
    ods select ParameterEstimates CollinDiag ResidualHistogram ResidualByPredicted
    RStudentByPredicted ObservedByPredicted CooksDPlot RStudentByLeverage
```

```
QQPlot
```

```
RFPlot ResidualPlot ObservedByPredicted;
model AvgTemperature=&_GLSMOD / ss1 collin;
run;
```

```
quit;
```

```
proc delete data=Work.reg_design;
run;
```

■ **/* ANCOVA (Analysis of covariance) test for variance*/**

```
title;  
ods noproctitle;  
ods graphics / imagemap=on;  
  
proc glm data=TEMP.CITY_TEMP plot(only)=(ancovaplot diagnostics);  
    class Region (ref='Africa');  
    model AvgTemperature=Region Year Year * Region / solution;  
    lsmeans Region / adjust=tukey pdiff alpha=.05 plots=(diffplot meanplot(cl) );  
quit;
```

■ **/* Logistic regression with a continuous predictor*/**

```
ods graphics on;  
proc logistic data= temp.city_temp alpha=0.05  
plots(only)=(effect oddsratio);  
model Region (event = '1')= AvgTemperature / clodds=both;  
title 'LOGISTIC MODEL (1)' : Temp_increase';  
run;
```